

Course Code: B23AD4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
GENERATIVE AI TECHNIQUES					
For AIDS					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			C O	KL	M
1.	a).	Define generative modeling in artificial intelligence.	1	1	2
	b).	List any two types of generative models.	1	1	2
	c).	What is a language model?	2	2	2
	d).	Name any two transformer-based text generation models	2	2	2
	e).	Expand the acronym GAN. What is CycleGAN used for?	3	2	2
	f).	Mention any two applications of image generation models.	3	2	2
	g).	List different OpenAI image models	4	1	2
	h).	What is prompt engineering?	4	2	2
	i).	Name any two open-source frameworks used in generative AI.	5	2	2
	j).	What is purpose of LangChain?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the difference between generative and discriminative modeling in AI with examples	1	3	5
	b).	Discuss the future directions and ethical challenges associated with generative AI.	1	2	5
3.	a).	Describe various types of generative models such as GANs, VAEs, and autoregressive models.	1	3	5
	b).	Describe the importance of generative modeling in AI and machine learning.	1	2	5
		UNIT-2			
4.	a).	Explain sequence-to-sequence models using LSTM with suitable architecture diagram	2	3	5
	b).	Explain the concept of attention mechanism and self-attention mathematically.	2	3	5
		OR			

5.	a).	Describe the Transformer architecture in detail including encoder, decoder, and attention mechanism	2	3	10
		UNIT-3			
6.	a).	Analyze any two computer vision architectures	3	3	5
	b).	Explain diffusion models and their stepwise generative process.	3	3	5
		OR			
7.	a).	Explain Variational Autoencoders (VAE) with loss function (ELBO) derivation.	3	3	5
	b).	Describe GAN architecture and training process	3	3	5
		UNIT-4			
8.	a).	Explain fine-tuning and plugin integration in ChatGPT systems.	4	3	5
	b).	Give the importance of prompt? Discuss prompt engineering process with examples.	4	3	5
		OR			
9.	a).	Discuss limitations of ChatGPT and future directions of generative AI systems.	4	3	5
	b).	Discuss various prompting techniques with examples	4	3	5
		UNIT-5			
10.	a).	Compare open-source and closed-source generative models with examples	5	3	5
	b).	Explain Hugging Face ecosystem and its features.	5	3	5
		OR			
11.	a).	Explain LangChain architecture and its components (agents, tools, memory).	5	3	5
	b).	Discuss Retrieval Augmented Generation (RAG) and its applications.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23HS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HUMAN RESOURCE & PROJECT MANAGEMENT					
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the role of HR manager.	1	1	2
	b).	Define recruitment.	1	1	2
	c).	List any two models of HR accounting.	2	1	2
	d).	Define career counseling.	2	1	2
	e).	Define Resource management.	3	1	2
	f).	State the 80-20 rule.	3	1	2
	g).	Define Brainstorming.	4	1	2
	h).	Write about Cash flow analysis.	4	1	2
	i).	List any two forms of project organization.	5	1	2
	j).	Define project control.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Interpret the nature and scope of Human Resource Management.	1	2	5
	b).	Discuss the emerging trends in HRM and their impact on organizations.	1	2	5
OR					
3.	a).	Explain the ethical aspects of HRM with suitable examples.	1	2	5
	b).	Describe the steps involved in the selection procedure.	1	2	5
UNIT-2					
4.	a).	Discuss the different methods of training used in organizations.	2	2	5
	b).	Explain the importance of performance appraisal in organizations.	2	2	5
OR					
5.	a).	Summarize the concept of career development and its importance.	2	2	5
	b).	Explain group interaction and its importance in HRD.	2	2	5
UNIT-3					
6.	a).	Explain the concept of Project Management and its importance.	3	2	5
	b).	Discuss the different types of projects with examples.	3	2	5
OR					
7.	a).	Explain the stages of the project life cycle.	3	2	5

	b).	Interpret the process of project selection.	3	2	5
		UNIT-4			
8.	a).	Describe the criteria used for project selection.	4	2	5
	b).	A company invests ₹2,00,000 in a project. The expected cash inflows are: Year 1: ₹50,000 Year 2: ₹60,000 Year 3: ₹70,000 Year 4: ₹80,000 Calculate the Payback Period.	4	2	5
		OR			
9.		A project requires ₹2,00,000 investment. It generates ₹80,000 annually for 4 years. Discount rate = 10% Calculate the NPV and state whether the project is acceptable.	4	2	10
		UNIT-5			
10.	a).	Interpret the process of project planning in detail.	5	2	5
	b).	Explain the human aspects of project management.	5	2	5
		OR			
11.	a).	Discuss the prerequisites for successful project implementation.	5	2	5
	b).	Summarize the importance of project review.	5	2	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CS4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE ARCHITECTURE & DESIGN PATTERNS					
(Common to CSE,AIDS, CSBS, CSD)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the role of Model–View–Controller (MVC) in design patterns.	1	2	2
	b).	Differentiate between cohesion and coupling.	1	2	2
	c).	Explain how actors are identified in use-case analysis.	2	2	2
	d).	Explain the concept of conceptual classes in system analysis.	2	2	2
	e).	Summarize the key idea behind the Adapter pattern.	3	2	2
	f).	Describe how the Proxy pattern controls access to another object.	3	2	2
	g).	Explain the role of the Model component in MVC architecture.	4	2	2
	h).	Explain how the undo operation can be supported in an interactive system.	4	2	2
	i).	Explain the concept of a distributed object system.	5	2	2
	j).	Explain the roles of client and server in a distributed application.	5	2	2
Estd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the essential elements used to document a design pattern.	1	2	5
	b).	Describe how design patterns help solve common design problems in software development.	1	2	5
		OR			
3.	a).	Describe the key concepts of object-oriented design.	1	2	5
	b).	Describe the drawbacks of the object-oriented paradigm.	1	2	5
		UNIT-2			
4.	a).	Use requirement gathering techniques to prepare a functional requirement specification for a student attendance management system.	2	3	5
	b).	Demonstrate how use-case analysis can be used to represent the functional requirements of a bank ATM system.	2	3	5
		OR			
5.	a).	Illustrate the process of identifying conceptual classes and relationships from a given problem description.	2	3	5
	b).	Apply domain knowledge to refine the analysis model of an e-commerce order processing system.	2	3	5

		UNIT-3			
6.	a).	Demonstrate how the Bridge pattern can be used to separate device abstraction from its implementation in a remote-control system.	3	3	5
	b).	Illustrate the use of the Decorator pattern to dynamically add additional features to a text editor component.	3	3	5
		OR			
7.	a).	Apply the Facade pattern to simplify the interaction between multiple subsystems in a banking application.	3	3	5
	b).	Demonstrate how the Flyweight pattern can be used to optimize memory usage in a large document editor with many repeated characters.	3	3	5
		UNIT-4			
8.	a).	Apply the MVC architectural pattern to design an interactive drawing application, identifying the model, view, and controller components.	4	3	5
	b).	Apply MVC principles to design the subsystems of an interactive online shopping interface.	4	3	5
		OR			
9.	a).	Demonstrate how a new feature can be added to an MVC-based system without modifying the entire architecture.	4	3	5
	b).	Illustrate how pattern-based solutions improve extensibility in an MVC-based interactive system.	4		5
		UNIT-5			
10.	a).	Demonstrate how Java Remote Method Invocation (RMI) can be used to enable communication between distributed objects in a networked application.	5	3	5
	b).	Demonstrate how RESTful web services can be used to provide data access in a student information system.	5	3	5
		OR			
11.	a).	Illustrate the process of implementing an object-oriented system on the web for an online shopping application.	5	3	5
	b).	Illustrate how an Enterprise Service Bus (ESB) can be used to integrate different enterprise applications in a distributed environment.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code:B23IT4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
BLOCKCHAIN TECHNOLOGY & APPLICATIONS					
(Common to IT, AIDS, CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			C O	KL	M
1.	a).	Define Blockchain and list its components.	1	1	2
	b).	What is decentralization in blockchain?	1	2	2
	c).	Define Smart Contract.	2	1	2
	d).	What are Oracles in blockchain?	2	2	2
	e).	What is a Private Blockchain?	3	1	2
	f).	Define Byzantine Fault Tolerance.	3	2	2
	g).	What is Blockchain scalability issue?	4	2	2
	h).	List any two applications of blockchain	4	2	2
	i).	What is Hyperledger Fabric?	5	1	2
	j).	List any two blockchain case study domains.	5	2	2
Estd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the origin and evolution of blockchain technology	1	2	5
	b).	Describe different types of blockchain and consensus mechanisms	1	2	5
		OR			
3.	a).	Explain the structure of a block in a blockchain with a neat diagram	1	2	5
	b).	Discuss cryptocurrency basics and types with suitable examples.	1	2	5
		UNIT-2			
4.		Explain Public Blockchain systems and the working of Bitcoin and Ethereum blockchain	2	3	10
		OR			
5.		Explain Smart Contracts, their characteristics, types, and role of oracles in blockchain	2	3	10
		UNIT-3			
6.	a).	Explain the architecture and characteristics of Private Blockchain systems	3	2	5

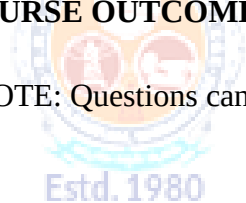
	b).	Describe Multichain and Byzantine Fault Tolerance in permissioned blockchain	3	3	5
		OR			
7.	a).	Explain Consortium Blockchain and its need in industry	3	2	5
	b).	Discuss Hyperledger, Ripple, and Corda platforms	3	2	5
		UNIT-4			
8.		Explain security and privacy challenges in blockchain along with identity management and smart contract security	4	3	10
		OR			
9.		Discuss various applications of blockchain in different domains and explain its limitations	4	2	10
		UNIT-5			
10.	a).	Explain blockchain case studies in retail and banking sectors	5	3	5
	b).	Describe blockchain implementation using Python and relevant packages	5	2	5
		OR			
11.	a).	Explain components of Hyperledger Fabric architecture.	5	2	5
	b).	Discuss chaincode and blockchain application development using Fabric Java SDK	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23AD4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Devops					
Common to AIDS & CSIT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the purpose of SDLC in software development?	1	2	2
	b).	How does DevOps improve the software development process?	1	2	2
	c).	What is the need for version control systems?	2	2	2
	d).	Why are branches used in Git?	2	2	2
	e).	What is the main goal of Continuous Integration?	3	2	2
	f).	What is the role of Jenkins in CI?	3	2	2
	g).	Why are Docker containers used in application deployment?	4	2	2
	h).	What is the purpose of Continuous Delivery?	4	2	2
	i).	What is the use of Ansible in DevOps?	5	2	2
	j).	What is the role of a Pod in Kubernetes?	5	2	2
Estd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the phases of SDLC with suitable examples.	1	2	5
	b).	Explain DevOps lifecycle and the role of automation in continuous improvement.	1	2	5
		OR			
3.	a).	Explain Agile Model and Compare Scrum and Kanban frameworks	1	3	5
	b).	Illustrate DevOps architecture and its key components.	1	3	5
		UNIT-2			
4.	a).	Explain the need for Source Code Management (SCM). Describe the evolution of SCM systems leading to Git.	2	2	5
	b).	Explain the Git workflow, clearly describing the roles of working directory, staging area, local repository, and remote repository.	2	2	5
		OR			
5.	a).	Explain Git branching and collaboration in team environments	2	2	5
	b).	Describe Unit Testing and explain the features of JUnit and NUnit	2	2	5

		frameworks used for automated testing.			
		UNIT-3			
6.	a).	Explain Build Automation and its importance in DevOps.	3	2	5
	b).	Explain Jenkins Architecture and Master–Slave model. Describe its working with a diagram.	3	2	5
		OR			
7.	a).	Explain Jenkins workflow with steps.	3	2	5
	b)	Compare Freestyle Projects and Jenkins Pipelines.	3	3	5
		UNIT-4			
8.	a).	Explain the importance of Continuous Delivery in DevOps.	4	2	5
	b)	Explain Dockerfile instructions and working of containers along with its lifecycle	4	2	5
		OR			
9.	a).	Explain Continuous Deployment flow with diagram.	4	2	5
	b)	Define Docker. Explain Docker Images, Containers, Commands and Dockerfile.	4	2	5
		UNIT-5			
10.	a).	Explain Ansible architecture and key features.	5	2	5
	b).	Explain Kubernetes architecture, namespaces, and resources.	5	3	5
		OR			
11.	a).	Explain Ansible roles and Jinja templating with examples.	5	2	5
	b).	Explain CI/CD in OpenShift. Describe BuildConfig (BC), DeploymentConfig (DC), and ConfigMaps	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23AD4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R23	
IV B.Tech. I Semester MODEL QUESTION PAPER					
ARCHITECTURE FOR MANAGEMENT OF LARGE DATASETS					
For AIDS					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Describe why HADOOP	1	2	2
	b).	Write Anatomy of file read	1	2	2
	c).	Define Big Data?	2	2	2
	d).	Describe HDFS	2	2	2
	e).	Define Hive	3	2	2
	f).	Write Hive Data types	3	2	2
	g).	Write Spark Applications	4	2	2
	h).	Describe Storm	4	2	2
	i).	Describe LLM	5	2	2
	j).	Write Applications of LLM	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	List out different challenges in Conventional Systems while handling big data	1	2	5
	b).	How Hadoop streaming is suited with text processing explain.	1	2	5
		OR			
3.	a).	Differentiate RDBBMS vs HADOOP	1	3	5
	b).	Explain Hadoop Configuration and Security	1	2	5
		UNIT-2			
4.	a).	Differentiate parallel and Distributed systems.	2	3	5
	b).	Explain HADOOP File System with neat diagram.	2	2	5
		OR			
5.	a).	Explain Symmetric Multi Processor (SMP)	2	2	5
	b).	Explain Shared Nothing Architecture	2	2	5
		UNIT-3			
6.	a).	Explain Hive Architecture with neat diagram	3	2	5

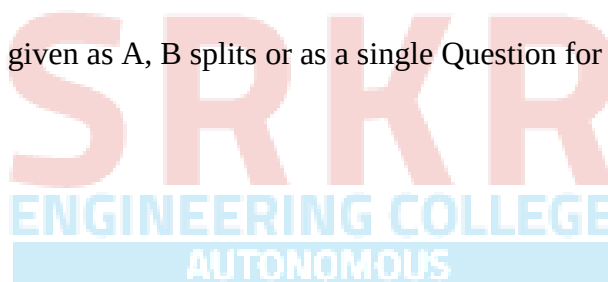
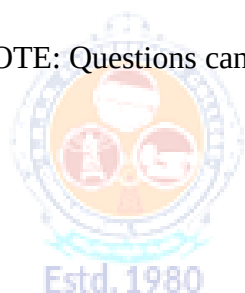
	b).	Explain User Defined functions (UDF) in Hive	3	2	5
		OR			
7.	a).	Explain Architecture of Graph.	3	2	5
	b).	Explain HiveQL and querying the Data.	3	2	5
		UNIT-4			
8.	a).	Explain the architecture of the storm.	4	2	5
	b).	Describe the anatomy of Spark Job Run.	4	2	5
		OR			
9.	a).	Explain jobs, stages and shared variables in Spark	4	2	5
	b).	Explain the process of installing spark	4	2	5
		UNIT-5			
10.	a).	Explain Classification of Digital Data.	5	2	5
	b).	Explain the stages of building and using LLMs.	5	2	5
		OR			
11.	a).	Explain transformer architecture.	5	2	5
	b).	Explain GPT architecture, Building a large language model.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23CS4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AGILE METHODOLOGIES					
(Common to CSE,AIIML, AIDS, CSBS, IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	What is meant by Agile manifesto	1	1	2
	b)	Define task board	1	1	2
	c)	Define self organizing team	2	1	2
	d)	Suggest appropriate agile communication principles for a software development team that is struggling with communication gaps and delayed deliveries in Ebook Reader project.	2	3	2
	e)	Mention the three main roles on a Scrum project.	3	1	2
	f)	How is Scrum both incremental and iterative?	3	2	2
	g)	Define pair programming	4	1	2
	h)	What is feedback loops and why do XP teams create them	4	2	2
	i)	Define Shuhari	5	1	2
	j)	What are the types of waste that create constraints in the workflow?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a)	Explain the core values of Agile manifesto and describe how they differ from traditional software development approaches	1	2	5
	b)	Analyze how a “fractured perspective” leads to project failure. How does Agile thinking help resolve this issue?	1	3	5
		OR			
3.	a)	Explain the idea of “Principles over Practices” in Agile. Why is focusing only on practices without understanding underlying principles often ineffective	1	2	5
	b)	Illustrate with an example how “Responding to Change over Following a Plan” benefits a software project.	1	3	5
		UNIT-2			
4.		Explain Agile principles and their importance in software development.	2	2	10
		OR			
5.	a)	Explain how effective communication contributes to better project	2	2	5

		delivery in software development.			
	b)	Explain how continuous improvement contributes to better project and team performance in Agile	2	2	5
		UNIT-3			
6.	a)	Explain the Scrum framework. Discuss the roles of Product Owner, Scrum master, and development team	3	2	5
	b)	Design a complete sprint planning process for a mobile application development project. Apply user stories, story points. Velocity estimation and task board management to create a comprehensive sprint plan.	3	3	5
		OR			
7.	a)	Explain the significance of Scrum values. Explain why Scrum practices alone are insufficient without adherence to these values.	3	2	5
	b)	A sprint fails because the Product Owner frequently changes priorities during sprint. Use Scrum planning rules to analyze the product owner support for sprint success without disrupting execution.	3	3	5
		UNIT-4			
8.		Explain the XP values and their role in shaping an effective development mindset	4	2	10
		OR			
9.	a)	Illustrate how continuous integration helps in maintaining software quality in XP	4	2	5
	b)	Explain how XP practices help teams embrace changing requirements effectively.	4	2	5
		UNIT-5			
10.	a)	Explain Lean thinking and identify different types of waste in software development	5	2	5
	b)	Analyse how limiting Work in Progress (WIP) improves flow in a Kanban system	5	3	5
		OR			
11.	a)	Explain Kanban principles to manage workflow in a software maintenance project	5	2	5
	b)	Evaluate the role of Agile Coach in helping organizations adopt agile practices successfully	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23AD4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R23	
IV B.Tech. I Semester MODEL QUESTION PAPER					
ROBOTIC PROCESS AUTOMATION					
For AIDS					
Time: 3hrs			Max. Marks :70		
Answer ONE Question from EACH UNIT					
All Questions carry equal marks					
Assume suitable data if necessary					
10 X 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Robotic Process Automation (RPA) and briefly explain its scope in modern business processes.	1	2	2
	b).	List any four key benefits of implementing RPA in an organization.	1	2	2
	c).	Define task recorder in UiPath and state its purpose.	2	2	2
	d).	Explain the concept of control flow in workflow design.	2	2	2
	e).	List different types of variables used in data manipulation.	3	2	2
	f).	Explain briefly the purpose of screen scraping and OCR.	3	2	2
	g).	Define assistant bots and their role in automation.	4	2	2
	h).	List common exceptions in RPA workflows.	4	2	2
	i).	Explain the role of the orchestration server in RPA.	5	2	2
	j).	List the steps involved in publishing a workflow.	5	2	2
5 X 10 = 50 Marks					
		UNIT – I	CO	KL	M
2.	a).	Elaborate the techniques of automation used in RPA.	1	2	5
	b).	Summarize the scope and benefits of RPA.	1	2	5
		OR			
3.	a).	What are the different roles of RPA platforms, explain with examples.	1	2	5
	b).	Explain the functionalities of Studio, Robot, and Orchestrator.	1	2	5
		UNIT – II			
4.	a).	Explain the use of Sequences, Activities, and Control Flow with a simple automation example.	2	2	5
	b).	Apply a task recorder to create a simple automation workflow.	2	3	5
		OR			
5.	a).	Summarize different types of loops used in workflows.	2	2	5
	b).	Apply control flow concepts to design a workflow with decision making.	2	3	5

		UNIT – III			
6.	a).	Explain collections and arguments in data handling.	3	2	5
	b).	Make use of data table operations to read and write CSV/Excel data.	3	3	5
		OR			
7.	a).	Explain mouse and keyboard activities used in UI automation.	3	2	5
	b).	Describe Screen Scraping and OCR techniques in UiPath.	3	2	5
		UNIT – IV			
8.	a).	Explain Assistant Bots and different types of triggers (Hotkey, Mouse, System, Image, Element).	4	2	5
	b).	Give an example of monitoring an email event or a keyboard event using triggers.	4	2	5
		OR			
9.	a).	Describe common exceptions in RPA and basic exception handling techniques.	4	2	5
	b).	Explain the role of debugging techniques and error reporting in RPA.	4	2	5
		UNIT – V			
10.	a).	Describe the process of publishing a workflow in UiPath.	5	2	5
	b).	Explain how to deploy bots using UiPath Orchestrator.	5	2	5
		OR			
11.	a).	Explain license management in RPA.	5	2	5
	b).	Explain the different methods for managing and updating bots.	5	2	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

IV B.Tech. I Semester MODEL QUESTION PAPER

REINFORCEMENT LEARNING

For AIDS

Time: 3 Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Identify the differences between supervised learning and Reinforcement Learning.	1	2	2
	b).	List any two limitations of Reinforcement Learning.	1	2	2
	c).	Define the goal of the bandit problem.	2	2	2
	d).	List any two advantages of optimistic initial values.	2	2	2
	e).	Define state and reward.	3	2	2
	f).	Define unified notation in Reinforcement Learning.	3	2	2
	g).	Define return in Monte Carlo methods.	4	2	2
	h).	Define incremental implementation.	4	2	2
	i).	State one achievement of TD-Gammon.	5	2	2
	j).	Define job-shop scheduling.	5	2	2

5 x 10 = 50 Marks

		UNIT-1			
2.	a).	Explain the history and evolution of Reinforcement Learning.	1	2	5
	b).	Explain the learning process in Tic-Tac-Toe using rewards.	1	2	5
		OR			
3.	a).	Explain the elements of Reinforcement Learning in detail.	1	2	5
	b).	Explain real-world applications of Reinforcement Learning	1	2	5
		UNIT-2			
4.	a).	Explain the n-Armed Bandit problem with an example.	2	2	10
		OR			
5.	a).	Discuss constant step-size methods for tracking nonstationary problems.	2	2	5
	b).	Explain Upper-Confidence Bound action selection method.	2	2	5
		UNIT-3			

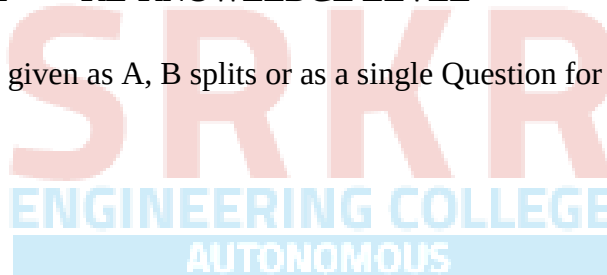
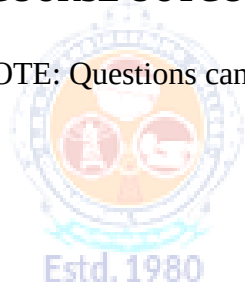
6.	a).	Explain the agent–environment interface with a neat diagram.	3	2	10
		OR			
7.	a).	Describe Markov Decision Processes in detail.	3	2	5
	b).	Differentiate value functions and optimal value functions.	3	2	5
		UNIT-4			
8.	a).	Discuss the advantages and limitations of Monte Carlo prediction.	4	2	5
	b).	Explain importance sampling on truncated returns.	4	2	5
		OR			
9.	a).	Differentiate on-policy and off-policy learning.	4	2	10
		UNIT-5			
10.	a).	Explain the Acrobot problem in Reinforcement Learning.	5	2	10
		OR			
11.	a).	Discuss optimization techniques used in elevator scheduling.	5	2	5
	b).	Explain Samuel’s Checkers Player and its contribution to Artificial Intelligence.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



IV B.Tech. I Semester MODEL QUESTION PAPER

HIGH PERFORMANCE COMPUTING

Common to AIML & AIDS

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE** Question from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define parallel computing.	1	2	2
	b).	What is implicit parallelism?	1	2	2
	c).	Define task decomposition.	2	2	2
	d).	What is load balancing?	2	2	2
	e).	What is broadcast communication?	3	2	2
	f).	Define reduction operation.	3	2	2
	g).	What is granularity in parallel systems?	4	2	2
	h).	Define scalability.	4	2	2
	i).	What is parallel quicksort?	5	2	2
	j).	What is CUDA?	5	2	2

5 x 10 = 50 Marks

		UNIT-1			
2.	a).	Explain the concept of motivating parallelism and its importance in modern computing.	1	2	5
	b).	Describe the trends in microprocessor and architectural design.	1	2	5
		OR			
3.	a).	Explain the physical organization of parallel computing platforms.	1	2	5
	b).	Discuss communication costs and scalable design principles in parallel systems.	1	2	5
		UNIT-2			
4.	a).	Explain different decomposition techniques used in parallel algorithm design.	2	2	5
	b).	Discuss the characteristics of tasks and their interactions with examples.	2	2	5
		OR			
5.	a).	Describe various mapping techniques for load balancing.	2	2	5

	b).	Explain methods for reducing interaction overheads in parallel programs.	2	2	5
		UNIT-3			
6.	a).	Explain one-to-all broadcast and all-to-one reduction operations.	3	2	5
	b).	Discuss all-reduce and prefix-sum operations with examples.	3	2	5
		OR			
7.	a).	Explain thread basics and synchronization mechanisms in shared memory systems.	3	2	5
	b).	Illustrate the OpenMP programming model with a simple example.	3	3	5
		UNIT-4			
8.	a).	Explain the sources of overhead in parallel programs.	4	2	5
	b).	Discuss various performance metrics for parallel systems.	4	2	5
		OR			
9.	a).	Explain the effect of granularity on performance and scalability.	4	2	5
	b).	Describe matrix-vector and matrix-matrix multiplication algorithms.	4	2	5
		UNIT-5			
10.	a).	Explain parallel bubble sort and its variants.	5	2	5
	b).	Describe the parallel quicksort algorithm.	5	2	5
		OR			
11.	a).	Explain the CUDA architecture and its components.	5	2	5
	b).	Describe CUDA programming: kernel execution, memory management, and synchronization.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks